

Concurrent File-Size Search

Objective

Develop a C program that performs a concurrent search for a file of a specific size. You will use multiple processes and signal handling to coordinate the search between two children and their parent.

Program Workflow

1. **Search Initialization:** The parent receives a target size and a list of files. It forks two children: **Child A** (searches the first half of the list) and **Child B** (searches the second half). If the number of files is odd, Child A holds more files.
2. **The "Winner" Scenario:**
 - If a child finds the file, it prints: I found the file at location [index] and exits.
 - The parent, upon detecting the child's exit, determines who won and prints: Parent: Child A (or B) found the file.
 - The parent then signals the surviving child that it has lost.
 - The losing child receives this signal and prints: I am child A (or B), and I received from my parent that I am the loser.
3. **The "Not Found" Scenario:**
 - After forking, the parent sets a 5-seconds time-out, if it receives no child exit during that time, the parent sends a signal to both children to terminate. Upon receiving this termination signal, each child prints: I am the child and I could not find the file.
 - The parent exits normally without printing.

Notes

- **Metadata Retrieval:** You must use the `stat()` system call to check file sizes. Do not attempt to open the files.
- **Signal Management:** Use signals wisely. No other IPC method should be used.
- **PID Tracking:** Ensure the parent stores the PIDs of both children immediately after `fork()` to identify them later.
- Every print should end with `\n`.
- Use `fflush(stdout)` after each print.
- The index printed by the child is 0-based

Examples

`./fsearch 400 f1.txt f2.txt f3.txt f4.txt`

I found the file at location 3.

Parent: Child B found the file.

I am child A, and I received from my parent that I am the loser.

`./fsearch 999 f1.txt f2.txt`

I am the child and I could not find the file.

I am the child and I could not find the file.